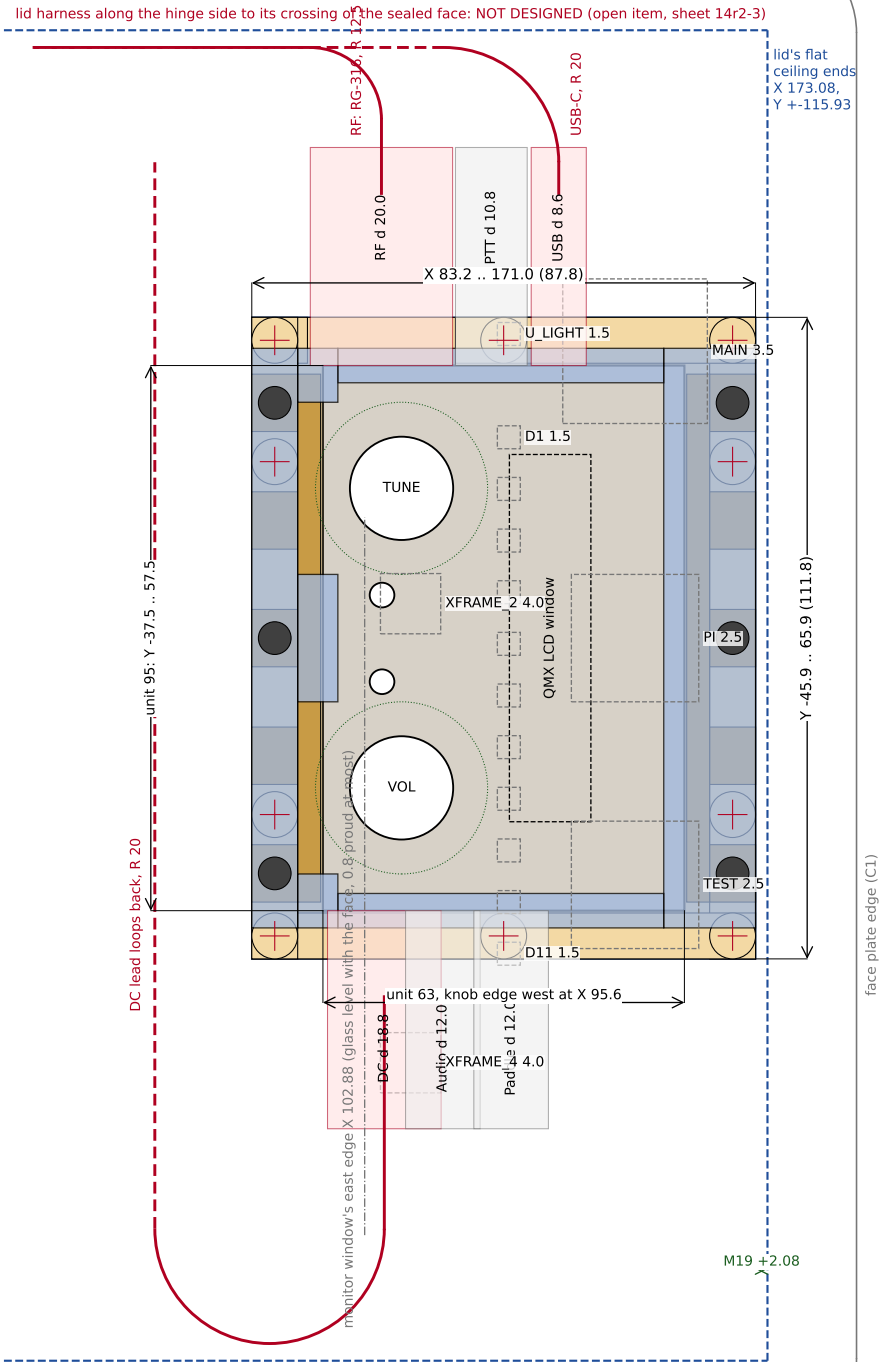
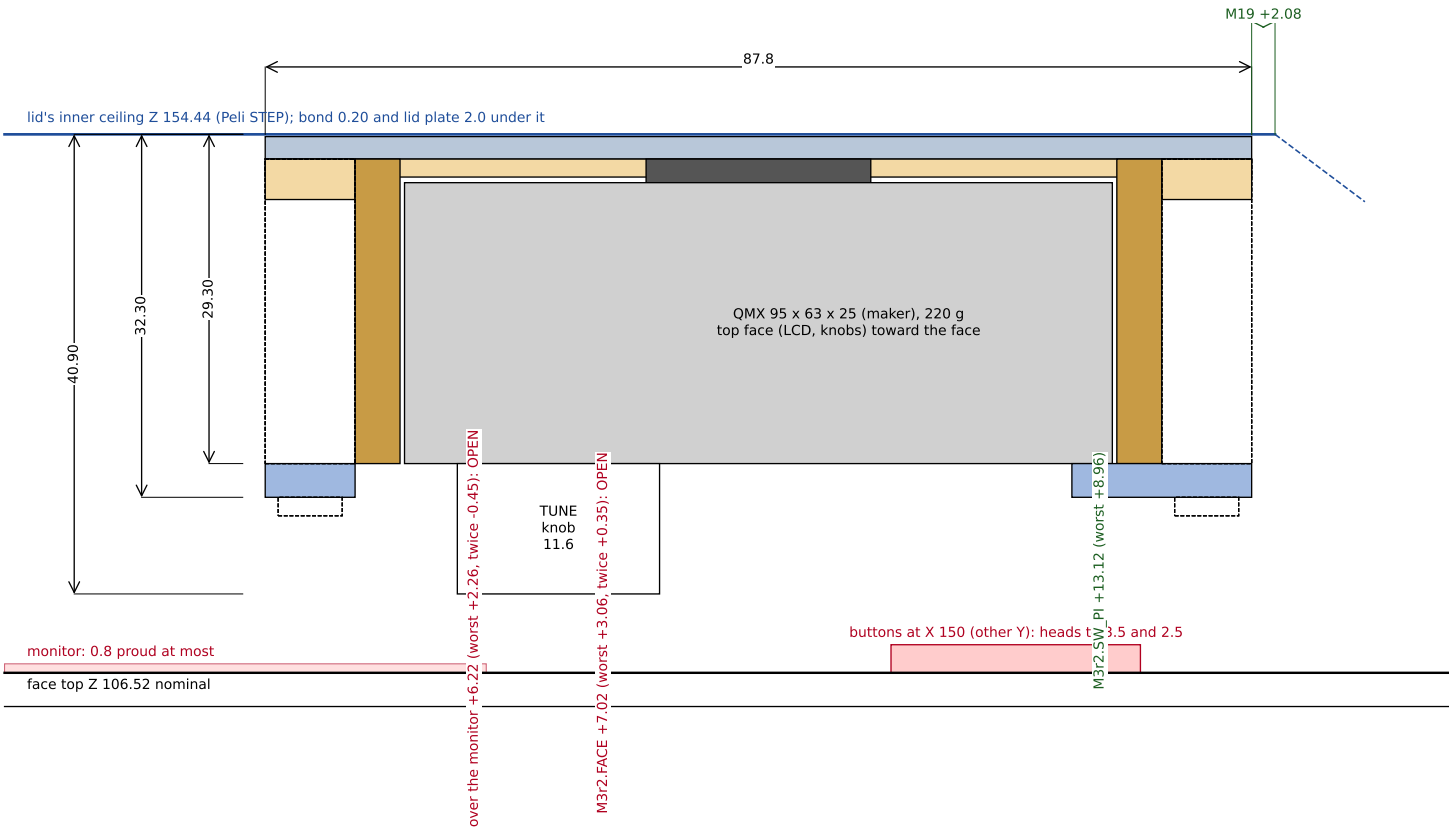


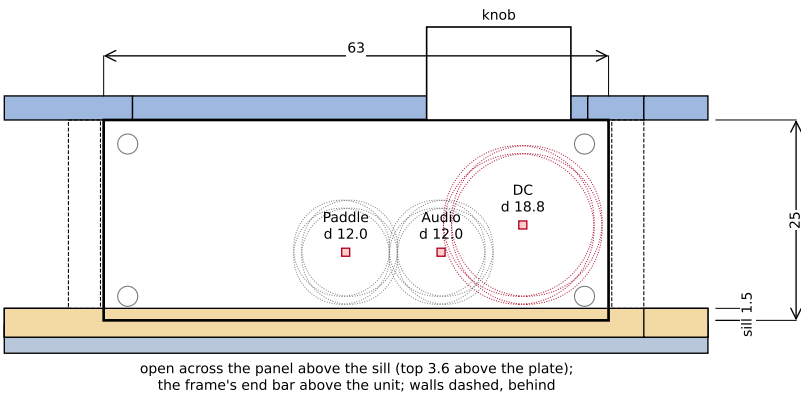
A. PLACE ON THE LID, in plan: case frame (X east, +Y to the hinge wall), lid closed, seen from above through the lid.
Blue outline: the lid plate (bonded); tan: the tray, brown its walls and posts; light blue: the retaining frame with its six screw heads; grey: the unit (right panel, RF PTT USB, toward the hinge; knob edge west); red crosses: the ten M3 in the plate;
red bands: the plugs of the permanent leads, grey bands: the operator's (unplugged before the lid closes); grey dashed: the face plate's parts under the set, each with its height above the face.



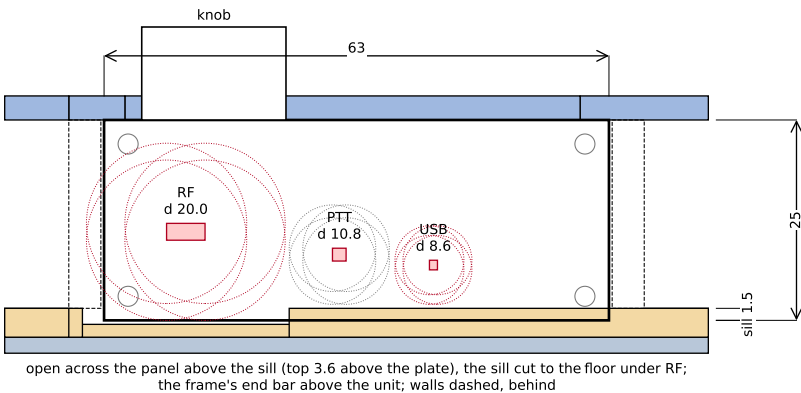
B. SECTION through the TUNE knob (Y 36.1), X east to the right, lid closed. Dashed: the posts and screw heads beside this plane.
Room from the face top to the ceiling 47.92 nominal, 44.39 at the worst, 42.11 with every unstated allowance twice (frame_seat.out M3); each row is taken from the deepest solid over the part, with the set's own allowances in it (the record, section B; sheet 14r2-3).



C1. LEFT panel (front end, toward case -Y), seen from outside: knob edge on the RIGHT



C2. RIGHT panel (hinge end, toward case +Y), seen from outside: knob edge on the LEFT



FACE ROOM, the rows that decide (lid_tray_gmx_r2_check.py, section B): nominal, worst, worst with unstated allowances twice																	
row and part	footprint, case X and Y	height	the deepest solid over it (of how many)					depth	nominal	worst	twice	verdict, min 1.00					
M3r2.XENARC WINDOW	X -102.9.. 102.9 Y	-94.0..	46.0	0.80	TUNE knob (22)				40.90	+6.22	+2.26	-0.45	OPEN, on an INFERRED height of the unit's				
M3r2.XFRAME 2	X 105.6.. 116.1 Y	10.8..	21.2	4.00	button actuator 2 (3)				31.30	+12.62	+8.66	+5.95	MET, on an INFERRED height of the unit's				
M3r2.SW MAIN	X 137.4.. 162.6 Y	47.4..	72.6	3.50	frame back rail and ledge (7)				32.30	+12.12	+7.96	+5.05	MET				
M3r2.SW PI	X 138.9.. 161.1 Y	-1.1..	21.1	2.50	frame back rail and ledge (4)				32.30	+13.12	+8.96	+6.05	MET				
M3r2.SW TEST	X 138.9.. 161.1 Y	-44.1..	-21.9	2.50	frame back rail and ledge (6)				32.30	+13.12	+8.96	+6.05	MET				
M3r2.D1	X 126.0.. 130.0 Y	43.0..	47.0	1.50	unit body (3)				29.30	+17.12	+13.16	+10.45	MET				
M3r2.D10	X 126.0.. 130.0 Y	-38.0..	-34.0	1.50	frame front-end bar (6)				32.30	+14.12	+9.96	+7.05	MET				
M3r2.FACE	X 100.3.. 118.3 Y	27.1..	45.1	0.00	TUNE knob (1)				40.90	+7.02	+3.06	+0.35	OPEN, on an INFERRED height of the unit's				
M19 (east edge at X 171.0 inside the flat ceiling): unchanged, nominal +2.08, worst +1.70, MET																	
every row of section B (the 18 face parts under the set) is on sheet 14r2-3; frame_seat.out's own M3 (+19.92, worst +16.39) is the r1 tray's and stays until frame_seat.py reads r2 (a draft)																	

MeshSat field kit V2, Peli 1450 case set

QMX lid tray r2: its place on the lid, the section with the lid closed, both end panels and their plugs

MESH SAT-1357, S-63, EQ-24. PROTOTYPE DESIGN: nothing made, printed, bought or fitted.
Session choices under the owner's standing rule of 26 Sep 2026. AI review only.

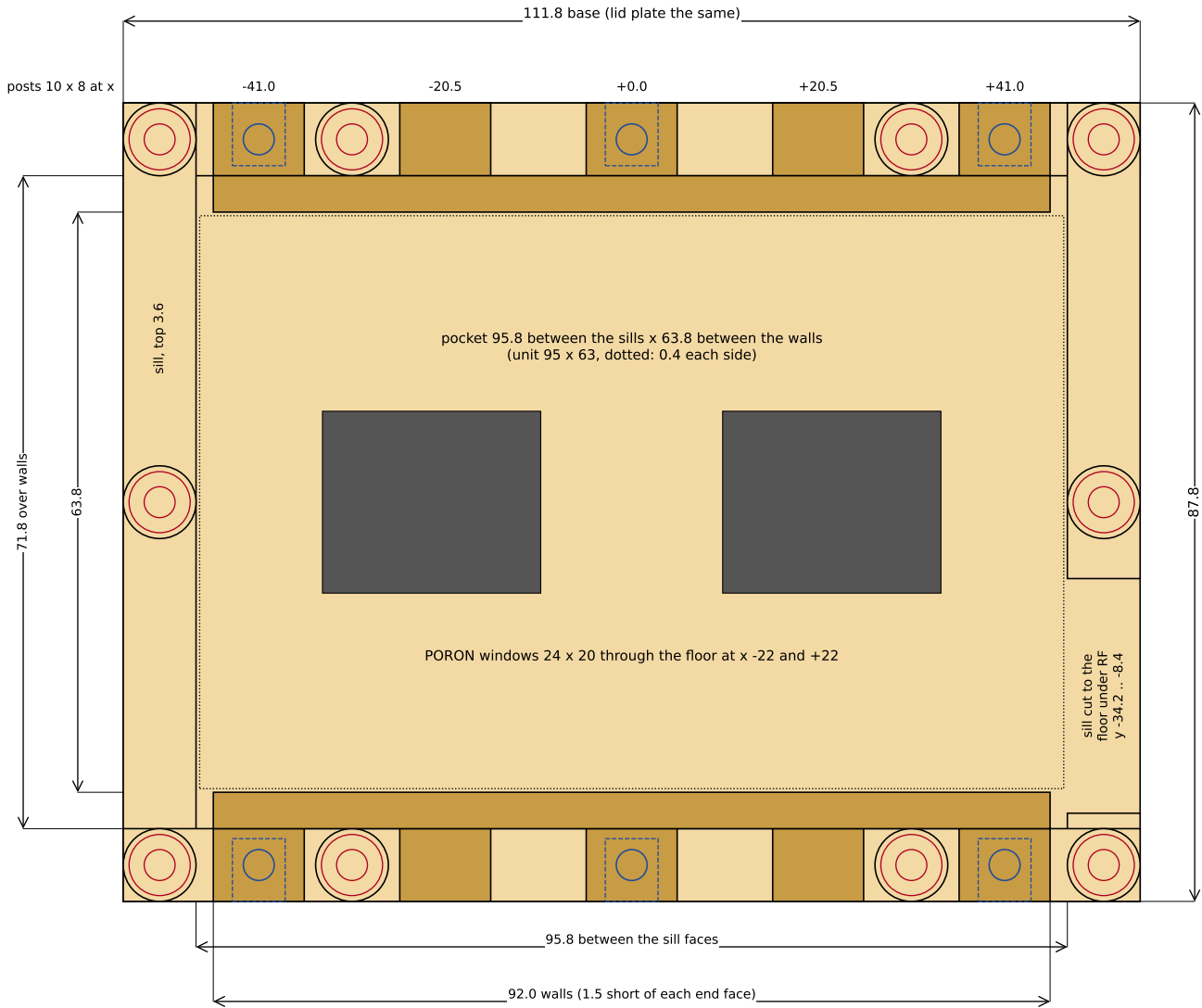
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210d43d126fa v2/cad/lid_tray_gmx_r2_check.py
3bdb88df9826 v2/ecad/tools/panel1450.py
a65c0792a7a0 v2/vendor/peli/1450/frame_seat.out
ec891c826be2 v2/vendor/grp-labs/gmx-figures.out
25f7aa7b99cc v2/vendor/materials/prusament-pc-blend-tds-v1.1-2022-02-16.pdf

Generated 2026-09-27 by v2/cad/lid_tray_gmx_r2_drawing.py from the files named left; regenerate with v2/cad/README.md. The editable sources govern; this sheet is a copy. Supersedes sheet 14 (lid-tray-gmx-drawing.pdf, the r1 tray of 9 Sep 2026).

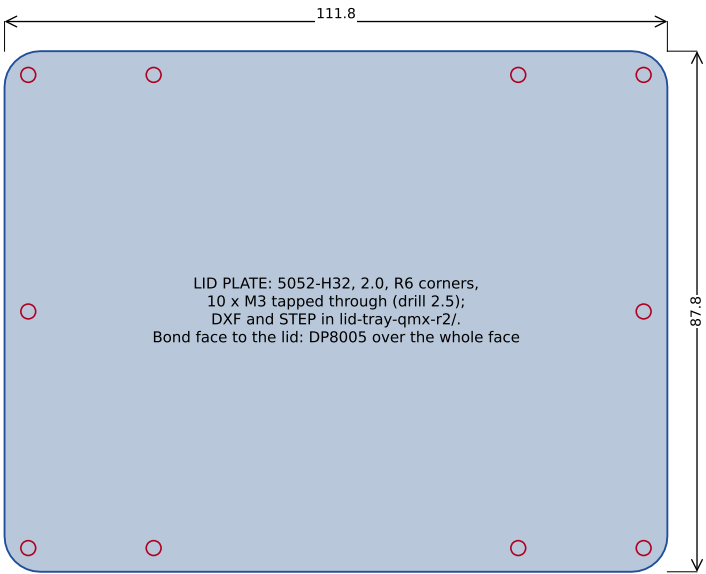
Sheet 14r2-1 of 3

A3, mm
not to scale for measurement:
dimensions govern

D. THE TRAY (print, Prusament PC Blend) in its own frame, seen from its open face: x toward the hinge (the unit's RIGHT panel), y east (the unit's back edge), z up from the lid plate. Brown: walls and posts to z 27.1; tan: floor 1.6, flanges and sills 3.6. Red: the ten countersunk M3 into the plate; blue: the six frame screws' holes and their nut slots (dashed, z 22.0 .. 24.1).



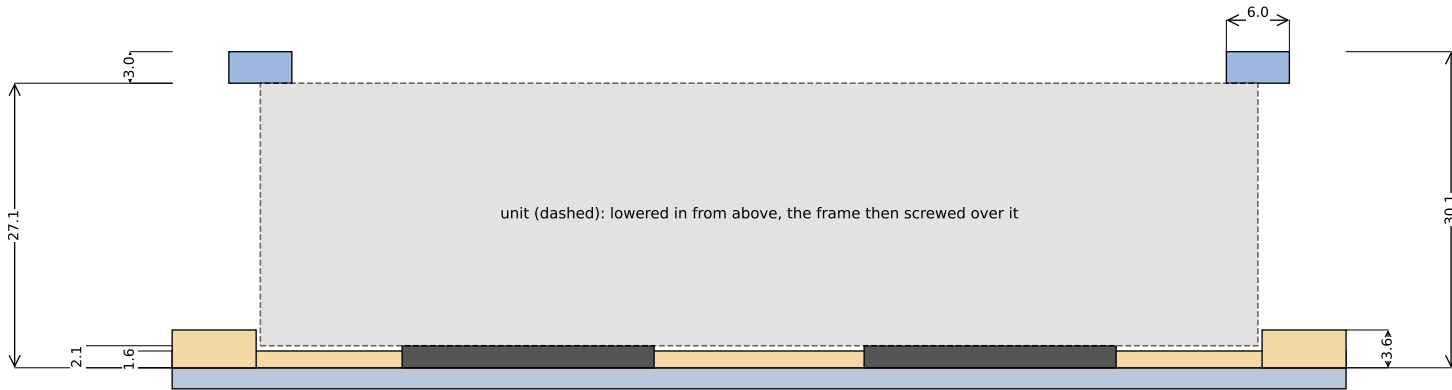
G. LID PLATE (plan, the tray's frame)



H0LES (part frame; the lid plate's tapped M3 at 1 to 10;
case X = 127.1 + y, Y = 10.0 + x)

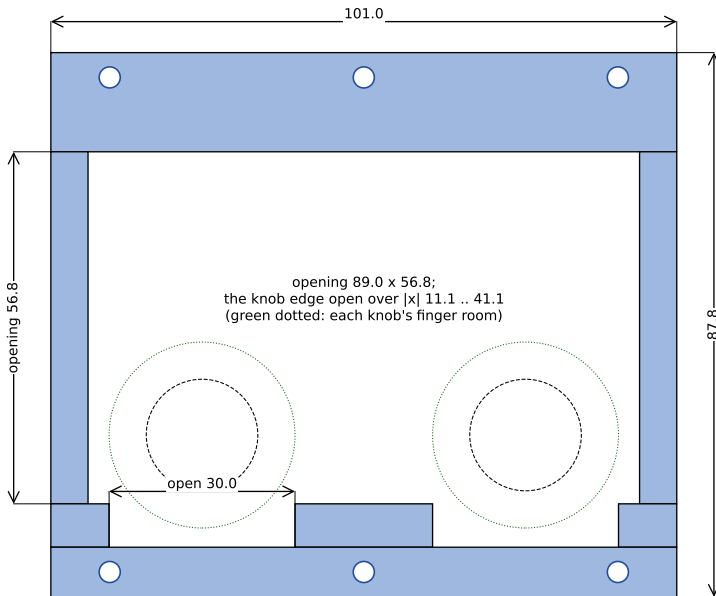
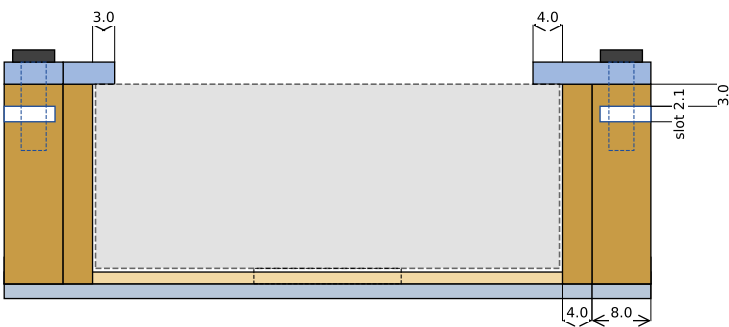
#	x	y	what
1	-30.75	-39.9	tray to plate, between the posts
2	-30.75	+39.9	tray to plate, between the posts
3	+30.75	-39.9	tray to plate, between the posts
4	+30.75	+39.9	tray to plate, between the posts
5	-51.90	-39.9	tray to plate, through a sill
6	-51.90	+0.0	tray to plate, through a sill
7	-51.90	+39.9	tray to plate, through a sill
8	+51.90	-39.9	tray to plate, through a sill
9	+51.90	+0.0	tray to plate, through a sill
10	+51.90	+39.9	tray to plate, through a sill
11	-41.00	-39.9	frame to post, nut slot z 22.0 .. 24.1
12	-41.00	+39.9	frame to post, nut slot z 22.0 .. 24.1
13	+0.00	-39.9	frame to post, nut slot z 22.0 .. 24.1
14	+0.00	+39.9	frame to post, nut slot z 22.0 .. 24.1
15	+41.00	-39.9	frame to post, nut slot z 22.0 .. 24.1
16	+41.00	+39.9	frame to post, nut slot z 22.0 .. 24.1

E1. SECTION ALONG THE LENGTH at y 0 (x to the right, z up from the lid plate): sills, pads, the frame's end bars



F. THE RETAINING FRAME (print, PC Blend, flat, 3.0 thick), in the tray's frame; six plain 3.4 holes

E2. SECTION ACROSS at x 0 (y to the right, the knob edge left), through two screwed posts: nut slots open outward (blue)



HARDWARE, MASSES, PRINT

HARDWARE. Ten M3 x 5 countersunk hex socket, ISO 10642 class, A2, each head flush in its 3.6 boss and its tip 0.6 short of the plate's back face. Six M3 x 10 button head hex socket, ISO 7380-1 class, A2 with six M3 square nut, DIN 562 class, A2 in the posts' slots; the heads stand 1.65 proud of the frame. Two pads, Rogers PORON 4701-30 Very Soft, 320 kg/m3, 3.18 mm (0.125 in), black (04), 24 x 20, in the floor windows on the plate. 3M DP8005 for the plate. **MASSES** (from the solids): tray 68.1 g, frame 12.2 g in PC Blend (1.22 g/cm3); lid plate 52.2 g in 5052 (2.68 g/cm3); the unit 220 g (maker). **PRINT.** Tray floor down, frame flat, Prusament PC Blend, 100 percent infill, 0.20 mm layers (the data sheet's specimen settings, which the retention figures assume). No part has an overhang; the only bridges are the six nut slots' roofs, 5.8 wide, which print without support. Printed dimensions +/-0.2 (INFERRED).

MeshSat field kit V2, Peli 1450 case set

QMX lid tray r2: the tray, the retaining frame and the lid plate, dimensioned

MESH SAT-1357, S-63, EQ-24. PROTOTYPE DESIGN: nothing made, printed, bought or fitted. Session choices under the owner's standing rule of 26 Sep 2026. AI review only.

Source: tree 1f26c1f2 (branch fnd/w5tray, from c23c5e76); inputs by sha256 (first 12):
f424d76a363f v2/cad/lid_tray_qmx_r2.py
210d43d126fa v2/cad/lid_tray_qmx_r2_check.py
3bdb88df9826 v2/ecad/tools/panel1450.py
a65c0792a7a0 v2/vendor/peli/1450/frame_seat.out
ec891c826be2 v2/vendor/grp-labs/qmx-figures.out
25f7aa7b99cc v2/vendor/materials/prusament-pc-blend-tds-v1.1-2022-02-16.pdf

Generated 2026-09-27 by v2/cad/lid_tray_qmx_r2_drawing.py from the files named left; regenerate with v2/cad/README.md. The editable sources govern; this sheet is a copy. Supersedes sheet 14 (lid-tray-qmx-drawing.pdf, the r1 tray of 9 Sep 2026).

Sheet 14r2-2 of 3

A3, mm
not to scale for measurement:
dimensions govern

NOTES

1. FITTING, in this order. (a) Bond the lid plate to the lid's unribbed inner face at the place of sheet 14r2-1 with DP8005 (clean, dry; the sheet asks no primer on polypropylene; fixture 2 h). (b) Screw the tray on with its ten M3 x 5, to the same torque as (f). (c) Lay the two pads in the floor windows, on the plate. (d) Slide a square nut into each of the six slots from outside, to the slot's end, and tape over the slot's mouth until its screw is in. (e) Do NOT fit the unit's four self-adhesive feet (the maker calls them optional). With the lid open and held level, and the knobs on, lower the unit into the pocket from above, top face out, knob edge west, right panel (RF, PTT, USB) toward the hinge: it stands on the pads, 1.08 above its place. (f) Lay the frame over it on the wall tops and drive the six M3 x 10 evenly, to 0.25 N m, until the frame sits on the wall tops all round; the pads are compressed by 1.08 (0.36 to 1.80 over the tolerances). (g) Plug the DC, RF and USB leads and tie each within 60 mm of its plug to a tie mount bonded to the lid, so no lead pulls on a jack. To take the unit out: the three plugs, the six frame screws, the frame, the unit.
2. THE WAY IN IS CHECKED (the record, section E): the unit at its largest, with its knobs, button actuators and jacks, keeps 0.30 to every solid of the tray on its way down (the least: unit body to the back wall), and the frame keeps 4.15 to everything that stands proud of the unit (the least: frame hinge-end bar to the RF jack, proud of the right panel); the stated minimum is 0.20. The first pass's tray, whose ledges and lintels were part of the tray, fails the same check, as it must.
3. OPERATION: the LCD, both knobs and both buttons face the operator with the lid open (the text then reads down the lid); the frame keeps 6 mm of finger room round each knob. The Paddle, Audio and PTT plugs are the operator's and come out before the lid closes.
4. OPEN, carried with this set (not closed by it): (a) THE LID HARNESS'S CROSSING OF THE SEALED FACE: how the DC lead, the USB-C lead and the RG-316 jumper get from the lid to boards B and A under the face plate, which has no pass-through, is designed nowhere; it blocks the harness picks, ASSEMBLY's lid harness steps and the closing of the lid on a wired QMX. (b) The lid plate's bond strength on the case's polypropylene, and after a thermal cycle (T8 pull test). (c) The peak E1 puts on the lid (E1 with an accelerometer on the lid); the set is designed to 100 g, an assumed peak. (d) The knob height (T9 with chalk on the knob tips, or the unit in hand): the knob rows are OPEN. (e) The lid's response in E2 and the pads' preload at their real compression. (f) The harness picks against the admitted plug bodies and R 20. (g) A threadlocker for the six frame screws and the ten tray screws that its maker states compatible with polycarbonate: none is named.

FACE ROOM, every row from the solid above the part (the record, section B): height above the face, depth from the lid ceiling, nominal, worst, worst with unstated allowances twice row and part									
footprint, case X and Y		height		the deepest solid over it (of how many)		depth	nominal	worst	twice verdict, min 1.00
M3r2.XENARC_WINDOW	X -102.9.. 102.9 Y -94.0.. 46.0	0.80	TUNE knob (22)	40.90	+6.22	+2.26	-0.45	OPEN, on an INFERRED height of the unit's	
M3r2.XFRAME_2	X 105.6.. 116.1 Y 10.8.. 21.2	4.00	button actuator 2 (3)	31.30	+12.62	+8.66	+5.95	MET, on an INFERRED height of the unit's	
M3r2.XFRAME_4	X 105.6.. 116.1 Y -69.2.. -58.8	4.00	DC plug, the body the set admits (1)	26.10	+17.82	+13.86	+11.15	MET	
M3r2.SW_MAIN	X 137.4.. 162.6 Y 47.4.. 72.6	3.50	frame back rail and ledge (7)	32.30	+12.12	+7.96	+5.05	MET	
M3r2.SW_PI	X 138.9.. 161.1 Y -1.1.. 21.1	2.50	frame back rail and ledge (4)	32.30	+13.12	+8.96	+6.05	MET	
M3r2.SW_TEST	X 138.9.. 161.1 Y -44.1.. -21.9	2.50	frame back rail and ledge (6)	32.30	+13.12	+8.96	+6.05	MET	
M3r2.D1	X 126.0.. 130.0 Y 43.0.. 47.0	1.50	unit body (3)	29.30	+17.12	+13.16	+10.45	MET	
M3r2.D2	X 126.0.. 130.0 Y 34.0.. 38.0	1.50	unit body (3)	29.30	+17.12	+13.16	+10.45	MET	
M3r2.D3	X 126.0.. 130.0 Y 25.0.. 29.0	1.50	unit body (3)	29.30	+17.12	+13.16	+10.45	MET	
M3r2.D4	X 126.0.. 130.0 Y 16.0.. 20.0	1.50	unit body (2)	29.30	+17.12	+13.16	+10.45	MET	
M3r2.D5	X 126.0.. 130.0 Y 7.0.. 11.0	1.50	unit body (2)	29.30	+17.12	+13.16	+10.45	MET	
M3r2.D6	X 126.0.. 130.0 Y -2.0.. 2.0	1.50	unit body (3)	29.30	+17.12	+13.16	+10.45	MET	
M3r2.D7	X 126.0.. 130.0 Y -11.0.. -7.0	1.50	unit body (3)	29.30	+17.12	+13.16	+10.45	MET	
M3r2.D8	X 126.0.. 130.0 Y -20.0.. -16.0	1.50	unit body (3)	29.30	+17.12	+13.16	+10.45	MET	
M3r2.D9	X 126.0.. 130.0 Y -29.0.. -25.0	1.50	unit body (2)	29.30	+17.12	+13.16	+10.45	MET	
M3r2.D10	X 126.0.. 130.0 Y -38.0.. -34.0	1.50	frame front-end bar (6)	32.30	+14.12	+9.96	+7.05	MET	
M3r2.D11	X 126.0.. 130.0 Y -47.0.. -43.0	1.50	front-end sill (left panel) (3)	5.80	+40.62	+36.66	+33.95	MET	
M3r2.U_LIGHT	X 126.0.. 130.0 Y 61.0.. 65.0	1.50	hinge-end sill, from the PTT jack east (3)	5.80	+40.62	+36.66	+33.95	MET	
M3r2.FACE	X 100.3.. 118.3 Y 27.1.. 45.1	0.00	TUNE knob (1)	40.90	+7.02	+3.06	+0.35	OPEN, on an INFERRED height of the unit's	

RETENTION (the record, section D): each link's capacity in N and in g on the unit, and its factor on the load case; the whole section, with E1, E2 and the sustained load, is in the record

LOAD CASE the set is designed to: 100 g on the unit and its plugs, 235 N, in each of the six directions, held still. It is the peak

CASE-MARGINS.md already assumes for the face under TEST-PLAN E1 (INFERRED: E1 sets 26 drops from 1.22 m, not a peak), not a measured one.

dir	link	N	g	factor
out	frame, back ledge bending at the wall's inner face (93.0 long, lever 2.70)	8990	3820	38.2 half the load on each side
out	frame, back rail bending between its screws and its ledge (arm 10.7, 27.1 wide at each of 3 screws)	1983	843	8.4 half the load on each side
out	back frame screws, 3, each holding 1019 N and prised by 3.67 on the post's outer edge	1664	707	7.1 half the load on each side
out	frame, knob-edge ledge bending at the wall's inner face (33.0 long, lever 2.20)	3915	1663	16.6 half the load on each side
out	frame, knob-edge rail bending between its screws and its ledge (arm 10.2, 26.1 wide at each of 3 screws)	2004	851	8.5 half the load on each side
out	knob-edge frame screws, 3, each holding 1019 N and prised by 3.55 on the post's outer edge	1723	732	7.3 half the load on each side
out	M3 threads in the lid plate, the four screws between the posts (in 1.4, tip 0.6 short of the bond)	3958	1682	16.8 INFERRED shear; the six end screws not counted
out	countersunk heads pulling through the 3.6 flange, the same four	6544	2781	27.8
out	tray flange bending between a post and the screw beside it (8.0 x 3.6, arm 5.25), eight places	2291	973	9.7 the floor not counted
out*	the lid plate's DP8005 bond on polypropylene: SCOPING ONLY (T-peel of 0.5 mm HDPE x perimeter 389)	613	168	no held figure bounds it
x	the left panel's bottom band bearing on its sill (1.1 x 61.0 at the loose corner)	4160	1768	17.7
x	the right panel's bottom band bearing on its sill (1.1 x 38.9 at the loose corner)	2653	1127	11.3 the sill cut away under the RF jack
x	the hinge-end sill sheared off the floor (8.0 x 52.3)	3975	1689	16.9 INFERRED shear
x	ten M3 through the flanges and both sills, bearing on the printed holes over 3.6	6696	2845	28.5 friction not counted
y	one wall pushed outward, bending across the layers at its root (load 13.0 above it, 4.5 post pitches)	2635	1120	11.2 the frame's tie not counted

one frame screw holds the least of:

- 1019 N the button head bearing on the 3.0 frame
- 1313 N the square nut bearing on its slot's 3.0 roof
- 1520 N the post and wall at the slot, in tension across the layers (80 mm2)
- 2264 N the screw, A2-70 (INFERRED class)

Governing link on the makers' typical figures: 707 g, 7.1 times the load case (direction out: back frame screws).

TOLERANCES AND STATUS

Dimensions in mm. Kit parts (this sheet): machined positions and outlines +-0.10; the 3.0 face plate +-0.13 and the 6.0 entry plates +-0.20 (EN 485-4 class, standard not held, INFERRED); the setting leg's pad height +-0.10; rebate depth and line +-0.10; spot-face floors +-0.10.

Peli: the 1450PF sheet 1453-314-000 rev A states +-0.76 on one-decimal and +-0.25 on two-decimal mm (VERIFIED); drawing 1451-931 states no tolerance ("typical industry-standard tolerances apply"), so every case allowance is the session's (INFERRED, UNSTATED).

Design basis only (v2/docs/CASE-MARGINS.md section 1, Verdicts): a MET row still meets its minimum with every unstated allowance taken twice, a sensitivity reading and not a bound on those allowances; an OPEN row rests on something no held source gives. Nominal CAD establishes no fit, seal or alignment; checks T1 to T11 (CASE-MARGINS.md section 5) do. Prototype: nothing on this sheet has been made, bought or fitted.

Jack places, the knobs and everything proud of the unit are SCALED or INFERRED from the maker's undimensioned figures and photographs (v2/vendor/qrp-labs/); the unit's 95 x 63 x 25 and 220 g are the maker's. Printed parts +-0.2 (FFF, INFERRED).

MeshSat field kit V2, Peli 1450 case set

QMX lid tray r2: fitting, operation, what stays open, and the record's face-room, fitting and retention rows

MESH SAT-1357, S-63, EQ-24. PROTOTYPE DESIGN: nothing made, printed, bought or fitted. Session choices under the owner's standing rule of 26 Sep 2026. AI review only.

Source: tree 1f26c1f2 (branch fnd/w5tray, from c23c5e76); inputs by sha256 (first 12):

- f424d76a363f v2/cad/lid_tray_gmx_r2.py
- 210d43d126fa v2/cad/lid_tray_gmx_r2_check.py
- 3bdb88df9826 v2/ecad/tools/panel1450.py
- a65c0792a7a0 v2/vendor/peli/1450/frame_seat.out
- ec891c826be2 v2/vendor/qrp-labs/gmx-figures.out
- 25f7aa7b99cc v2/vendor/materials/prusament-pc-blend-tds-v1.1-2022-02-16.pdf

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Sheet 14r2-3 of 3

A3, mm not to scale for measurement: dimensions govern